

Chuqur o'rganish usullari

3-mavzu. Konvolyutsion neyron tarmoqlar (CNN) — Practice

Mavzu: konvolyutsiya (kross-korrelyatsiya), stride, padding, dilation, chiqish o'lchami, parametrlar va MAC, 1×1 va depthwise separable konvolyutsiya, pooling (MaxPool2d, AvgPool2d, GAP), qabul maydoni, konvolyutsiya orqali backpropagation, im2col, transposed konvolyutsiya, batch normalization, residual bog'lanish, augmentatsiya va transfer learning.

Maqsad: (1) konvolyutsiya, pooling va BN ni qo'lda hisoblash; (2) o'lcham, parametr va qabul maydoni formulalarini qo'llash va isbotlash; (3) Conv2d, pooling, BN ning oldinga va orqaga yurishini NumPy'da yozib, PyTorch bilan solishtirish; (4) kichik CNN larni real tasvirlarda o'qitish va tahlil qilish.

Baholash: har bir topshiriq 1 ball, jami 30 ball. Hisob topshiriqlarida javob 4 ta o'nlik xona aniqlikda; kod topshiriqlarida (17–30) kod xatosiz ishlashi, chop etilgan natija ko'rsatilgan formatda bo'lishi va 1–2 jumlati xulosa talab qilinadi.

Muhit: Python 3.11/3.12, `pip install -r requirements.txt` (versiyalar qat'iy: numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0, scikit-image 0.26.0, torch 2.14.0, faqat CPU). Ma'lumotlar: `sklearn.datasets.load_digits` va `skimage.data` (paket ichida, internet shart emas). Kod bitta `solutions.py` faylida `task_17()`, ..., `task_30()` funksiyalari sifatida yoziladi. Hujjatlar: <https://pytorch.org/docs/stable/generated/torch.nn.Conv2d.html>, <https://pytorch.org/docs/stable/generated/torch.nn.BatchNorm2d.html>, <https://scikit-image.org/docs/stable/api/skimage.data.html>.

A. Qo'lda hisoblash, isbot va tushunchalar (1–16)

“Konvolyutsiya” deganda Conv2d dagidek kross-korrelyatsiya tushuniladi: $Y_{i,j} = b + \sum_{u,v} K_{u,v} X_{i+u,j+v}$ (indekslar 0 dan).

- (hisob) $X = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & -1 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 0 & -1 & 0 \end{pmatrix}$, $K = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$, $b = 0$. (a) $s = 1$, $p = 0$ dagi Y ; (b) $s = 2$ dagi Y ; (c)

matematik (ag'darilgan yadroli) konvolyutsiya $s = 1$ da nima beradi?

- (hisob) Ikki kanalli kirish $X_1 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$, $X_2 = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$, bitta filtr $\mathcal{W}_1 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, $\mathcal{W}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

$b = -1$. Chiqish xaritasini va qatlam parametrlari sonini toping. $C_{\text{out}} = 8$ bo'lsa-chi?

- (isbot, hisob) $H_{\text{out}} = \lfloor (H + 2p - d(K - 1) - 1) / s \rfloor + 1$ ni isbotlang. Hisoblang: $(H, K, p, s, d) = (64, 3, 1, 2, 1)$, $(28, 3, 0, 1, 1)$, $(32, 5, 2, 1, 1)$, $(15, 3, 1, 2, 1)$, $(32, 3, 2, 1, 2)$, $(100, 7, 3, 4, 1)$, $(28, 3, 0, 2, 2)$. $3 \times 64 \times 64$ kirish `Conv2d(3,32,5) → MaxPool2d(2) → Conv2d(32,64,3,stride=2,padding=1) → MaxPool2d(2) → Flatten` dan o'tsa, keyingi `Linear` ning kirish o'lchami qancha?

- (hisob) (a) `Conv2d(3,16,5,padding=2)` ning 32×32 tasvirdagi parametrlari va MAC soni. (b) Xuddi shu kirish $(3 \cdot 32 \cdot 32)$ va chiqish $(16 \cdot 32 \cdot 32)$ o'lchamli to'liq bog'langan qatlam parametrlari. (c) 1×1 konvolyutsiya $256 \rightarrow 64$ parametrlari. (d) $128 \rightarrow 256$ uchun oddiy 3×3 va depthwise separable (3×3 , `groups=128`, so'ng 1×1) parametrlari, ularning nisbati va MAC nisbati $\frac{1}{C_{\text{out}}} + \frac{1}{K^2}$.

- (isbot, hisob) $s = 1$ konvolyutsiya siljishga ekvivalent ekanini isbotlang: $\mathcal{C}(T_a X) = T_a \mathcal{C}(X)$. 1D misol: $\mathbf{x} = (0, 0, 1, 2, 0, 0)$, $\mathbf{k} = (1, -1)$, $p = 0$; $\mathbf{x}$ ni bir pozitsiya o'ngga siljitib (`roll`) ikkala chiqishni solishtiring. So'ng oynasi 2, qadami 2 bo'lgan max-pooling uchun xuddi shu tajribani bajaring. Xulosa?

- (hisob) Qabul maydoni: $r_l = r_{l-1} + (k_l - 1)d_l j_{l-1}$, $j_l = j_{l-1}s_l$, $r_0 = j_0 = 1$. Tarmoq: `conv 3 × 3` ($s = 1$) → `conv 3 × 3` ($s = 2$) → `conv 3 × 3` ($d = 2$) → `maxpool 2 × 2` ($s = 2$) → `conv 3 × 3`. Har qatlamdan keyingi r_l , j_l ni toping. $C = 64$ kanalda uchta 3×3 qatlam va bitta 7×7 qatlam og'irliklari sonini (siljishsiz) solishtiring.

- (hisob) $X = \begin{pmatrix} 1 & 5 & 2 & 0 \\ 3 & 2 & 4 & 6 \\ 0 & 1 & 3 & 2 \\ 7 & 2 & 1 & 0 \end{pmatrix}$, 2×2 , $s = 2$. Max-pooling va average pooling chiqishlarini, so'ng yuqoridan

kelgan $\bar{Y} = \begin{pmatrix} 1 & -1 \\ 2 & 0,5 \end{pmatrix}$ uchun ikkala holatda $\bar{X}$ ni toping.

- (isbot, hisob) $s = 1$, $p = 0$ uchun $\frac{\partial L}{\partial K_{u,v}} = \sum_{i,j} \delta_{i,j} X_{i+u,j+v}$, $\frac{\partial L}{\partial b} = \sum_{i,j} \delta_{i,j}$, $\frac{\partial L}{\partial X_{a,b}} = \sum_{u,v} \delta_{a-u,b-v} K_{u,v}$ ni isbotlang. Hisoblang: $X = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$, $K = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}$, $\delta = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$.

9. (*hisob, tushuncha*) Transposed konvolyutsiya: $H_{\text{out}} = (H-1)s - 2p + d(K-1) + p_{\text{out}} + 1$. ($H, K, s, p, p_{\text{out}} = (7, 3, 2, 1, 1)$ va $(16, 4, 2, 1, 0)$ uchun H_{out} ni toping. `ConvTranspose2d(64, 32, 4, stride=2, padding=1)` parametrlari soni? Nima uchun X bo'yicha gradient transposed konvolyutsiya bilan hisoblanadi?
10. (*hisob*) BatchNorm, bitta kanal qiymatlari $(2, 4, 4, 6, 9)$, $\gamma = 2$, $\beta = 1$, $\epsilon = 0$. μ , σ^2 ($1/N$), $\hat{x}$, y ni toping. PyTorch'da boshlang'ich `running_mean = 0`, `running_var = 1`, `momentum = 0,1` va `running` dispersiya *siljimagan* ($1/(N-1)$) baho bilan yangilanadi: bir qadamdan keyingi qiymatlar va `eval()` rejimida $x = 9$ uchun chiqish ($\epsilon = 10^{-5}$) qancha?
11. (*isbot*) BN orqali gradient $\frac{\partial L}{\partial x_k} = \frac{\gamma}{N_s} (N \bar{y}_k - \sum_j \bar{y}_j - \hat{x}_k \sum_j \bar{y}_j \hat{x}_j)$ formulasidan $\sum_k \partial L / \partial x_k = 0$ va $\sum_k \hat{x}_k \partial L / \partial x_k = 0$ ekanini isbotlang ($\sum \hat{x}_k = 0$, $\sum \hat{x}_k^2 = N$ dan foydalaning). Bu tengliklarning ma'nosi nima?
12. (*hisob, tushuncha*) Residual blok $y = x + F(x)$. Skalyar holatda har blokda $\partial F / \partial x = 0,2$ bo'lsa, 10 blokdan keyin oddiy tarmoq va residual tarmoqdagi gradient ko'paytuvchisini toping. `Conv(32, 32, 3, padding=1, bias=False)`–BN–ReLU–Conv–BN dan iborat blok parametrlari va `Conv2d(32, 64, 1, stride=2, bias=False)` proyeksiya parametrlari soni?
13. (*hisob*) $3 \times 32 \times 32$ kirish, tarmoq: `Conv2d(3, 16, 3, padding=1)`–ReLU–MaxPool2d(2)–`Conv2d(16, 32, 3, padding=1)`–ReLU–MaxPool2d(2)–`Conv2d(32, 64, 3, padding=1)`–ReLU–`AdaptiveAvgPool2d(1)`–`Flatten`–`Linear(64, 10)`. Har bir qatlamdan keyingi o'lcham va parametrlar, jami parametrlar. GAP o'rniga `Flatten`+`Linear` qo'yilsa, bosh nechta parametrga ega bo'ladi?
14. (*amaliy holat: sifat nazorati*) 4×4 tasvirning yuqori ikki qatori 0 (qora), pastki ikki qatori 1 (oq). $G_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$ va $G_y = G_x^T$ bilan ($p = 0$) javoblarni, gradient kattaligi $\sqrt{G_x^2 + G_y^2}$ va yo'nalishini toping. Qanday qirra topildi? Ag'darilgan konvolyutsiya ishorani qanday o'zgartiradi?
15. (*tushuncha*) (a) ResNet-18 da barcha qatlamlar muzlatilib, `fc 512 → 3` ga almashtirilsa, nechta parametr o'qitiladi? (b) Quyidagilarning har biri nima uchun xato va qanday tuzatiladi: (i) `img` (`NumPy`, $H \times W \times C$) to'g'ridan-to'g'ri `Conv2d` ga berildi; (ii) BN li model `model.eval()` siz baholandi; (iii) raqamlar (0–9) uchun `RandomHorizontalFlip` augmentatsiyasi; (iv) augmentatsiya test to'plamiga ham qo'llandi. (c) Nega CNN jadval (tabular) ma'lumotlarida odatda foydasiz?
16. (*amaliy holat: sensor signali*) Harorat sensori signali $x = (1, 2, 3, 10, 3, 2, 1)$ ($p = 0, 1D$). (a) Sirg'aluvchi o'rtacha $k = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$; (b) farq $k = (-1, 0, 1)$; (c) 3 lik oynali median. Chiqishlarni toping. Qaysi filtr "sakrash"ni (anomaliyani) aniqlaydi, qaysi biri uni yo'qotadi? Median filtr nima uchun `Conv1d` bilan ifodalanmaydi?

B. Dasturlash (17–30)

Umumiy qoidalar. `import numpy as np, torch, torch.nn as nn, torch.nn.functional as F;` har bir model yaratilishidan oldin `torch.manual_seed(42);` boshida `torch.set_num_threads(1)` va `torch.use_deterministic_algorithms(True)`. **Digits** (26–27, 30): `X, y = load_digits(return_X_y=True); X = X.reshape(-1, 1, 8, 8) / 16.0;` `train_test_split(X, y, test_size=0.3, stratify=y, random_state=42);` `float32`. **O'qitish funksiyasi** `fit(model, X, y, epochs=30, lr=3e-3, augment=False, params=None):` `Adam`, `gen = torch.Generator().manual_seed(42)`, `DataLoader(TensorDataset(X, y), batch_size=32, shuffle=True, generator=gen)`, `F.cross_entropy`, oxirida `model.eval()`. **Digits CNN:** `Conv2d(1, 16, 3, padding=1)`–ReLU–MaxPool2d(2)–`Conv2d(16, 32, 3, padding=1)`–ReLU–MaxPool2d(2)–`Flatten`–`Linear(128, n_out)`; **Digits MLP:** `Flatten`–`Linear(64, 64)`–ReLU–`Linear(64, 10)`. Kod `python -W error::DeprecationWarning solutions.py` bilan xatosiz ishlashi kerak.

17. (*sodda konvolyutsiya*) `out_size(h, k, p=0, s=1, d=1)` va `conv2d_naive(X, W, b, s=1, p=0, d=1)` ($X: (C_{\text{in}}, H, W)$, $W: (C_{\text{out}}, C_{\text{in}}, K, K)$, sikllar bilan) yozing. `rng = np.random.default_rng(42); X = rng.normal(size=(3, 9, 9)); W = rng.normal(size=(4, 3, 3, 3)); b = rng.normal(size=4).` ($s, p, d \in \{(1, 0, 1), (1, 1, 1), (2, 1, 1), (1, 2, 2), (3, 0, 1)\}$ uchun chiqish shakli va `F.conv2d` bilan mosligini (`max|diff| < 1e-12`) chop eting.
18. (*o'lcham formulalari*) 3-topshiriqdagi 7 ta holat uchun `out_size` va `nn.Conv2d(1, 1, K, stride, padding, dilation)(torch.zeros(1, 1, H, H))` natijasini jadval qilib chop eting; `MaxPool2d(3, stride=2, padding=1)` (112) va `tconv_size(h, k, s, p, op)` ni 9-topshiriqdagi ikki holat uchun `nn.ConvTranspose2d` bilan tekshiring.
19. (*im2col*) `im2col(X, K, s=1, p=0) -> (cols, Ho, Wo):` har bir oyna (`X[:, i:i+K, j:j+K].reshape(-1)`) — bitta ustun, ustunlar qatorlar bo'yicha tartibda. `rng = np.random.default_rng(42); X = rng.normal(size=(3, 8, 8)); W = rng.normal(size=(5, 3, 3, 3)); b = rng.normal(size=5); p = 1.` $Y = W_{\text{row}} X_{\text{col}} + b$ ni `F.conv2d` bilan, `cols` ni `F.unfold` bilan solishtiring; shakllarni chop eting.

20. (*konvolyutsiya orqali backprop*) `conv_backward(X, W, delta) -> (dW, db, dX)` ($s = 1, p = 0$; 8-topshiriq formulalari, (u, v) bo'yicha sikl va `np.einsum`). 8-topshiriq misolini (δ bilan) chop eting. So'ng `rng = np.random.default_rng(42)`; `X = rng.normal(size=(2, 5, 5))`; `W = rng.normal(size=(3, 2, 3, 3))`; `delta = rng.normal(size=(3, 3, 3))` uchun $\sum(Y \odot \delta)$ ning autograd gradientlari bilan solishtiring.
21. (*pooling*) `maxpool2x2(X) -> (Y, mask)` va `maxpool2x2_backward(dY, mask)` (`reshape(C, H//2, 2, W//2, 2)` va `np.repeat`). `rng = np.random.default_rng(42)`; `X = rng.normal(size=(2, 4, 6))`; `dY = rng.normal(size=(2, 2, 3))`. `F.max_pool2d` va autograd bilan solishtiring, noldan farqli gradientlar sonini chop eting; average pooling uchun $\bar{X} = \bar{Y}/4$ ni tekshiring.
22. (*BatchNorm2d*) `torch.manual_seed(42)`; `x = torch.randn(8, 3, 4, 4, dtype=torch.float64, requires_grad=True)`; `g = torch.randn(8, 3, 4, 4, dtype=torch.float64)`; `bn = nn.BatchNorm2d(3, dtype=torch.float64)`, $\gamma = (1; 2; 0.5)$, $\beta = (0; 1; -1)$. Train rejimida `(bn(x) * g).sum().backward()`. Qo'lda: forward, 11-topshiriqdagi backward formulasi, kanal bo'yicha $\sum \partial L / \partial x = 0$, running statistikalar va `eval()` chiqishi. Har bir tekshiruvni `True/False` va $\max |y_{\text{train}} - y_{\text{eval}}|$ (.4f) ni chop eting.
23. (*amaliy holat: qirralarni aniqlash*) `img = skimage.data.camera() / 255` (`float64, 512 × 512`). G_x, G_y (14-topshiriq) ni bitta `F.conv2d` (2 ta chiqish kanali, $p = 0$) bilan qo'llang; natijani `scipy.ndimage.sobel(img, axis=1)` va `axis=0` ning ichki qismi (`[1:-1, 1:-1]`) bilan solishtiring. Chiqish shakli, $|G|$ ning o'rtachasi, maksimumi va $|G| > 0.5$ pikseller ulushini (.4f) chop eting. Hujjat: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.sobel.html>.
24. (*ekvivariantlik*) `torch.manual_seed(42)`; `x = torch.randn(1, 1, 16, 16)`; siljish T — `torch.roll(t, (2, 3), dims=(2, 3))`. `padding_mode ∈ {circular, zeros}` uchun (har biridan oldin `torch.manual_seed(42)`) `nn.Conv2d(1, 4, 3, padding=1, padding_mode=...)`: $\max |C(Tx) - TC(x)|$ ni butun xaritada va ichki qismida (`[..., 4:12, 4:12]`) chop eting (.2e). `MaxPool2d(2)` uchun 2 va 1 ga siljitishda (chiqishni mos ravishda 1 va 0 ga) farqni chop eting.
25. (*o'lcham yo'li va parametrlar*) 13-topshiriqdagi tarmoqni quring va $1 \times 3 \times 32 \times 32$ nol tensorni qatlam-qatlam o'tkazib, har qatlam nomi, chiqish shakli va parametrlar sonini chop eting; jami, `Flatten` bosh varianti va 4(d)-topshiriqdagi ikki qatlam (`groups=128`) parametrlarini tekshiring.
26. (*amaliy holat: pochta indeksleri — raqamlar*) Digits CNN (`n_out=10`) va Digits MLP ni `fit` bilan o'qiting (har biridan oldin `torch.manual_seed(42)`). Parametrlar soni, train va test accuracy (.4f) ni chop eting.
27. (*siljishga chidamlilik va augmentatsiya*) `shift_images(X, dx, dy=0)` yozing: tasvirni dx piksel o'ngga, dy pastga siljitadi, bo'shagan joy 0. Uch model (har biridan oldin `torch.manual_seed(42)`): MLP, CNN, CNN+aug. Augmentatsiya: har bir paketda `sh = torch.randint(-1, 2, (len(xb), 2), generator=gen)` va har bir namuna `shift_images(x, sh[k,0], sh[k,1])` bilan siljiriladi. Test tasvirlarini 0, 1, 2 piksel o'ngga siljitib accuracy jadvalini chop eting va natijani izohlang.
28. (*residual va gradient oqimi*) `torch.manual_seed(42)`; `x = torch.randn(8, 16, 16, 16)`; `y = torch.randint(0, 10, (8,))`. Ikki tana (har biridan oldin `torch.manual_seed(42)`): 20 ta `PlainBlock`: `ReLU(conv(x))` va 20 ta `ResBlock`: $x + \text{conv}_2(\text{ReLU}(\text{conv}_1(x)))$ (`Conv2d(16,16,3,padding=1)`, standart initialization), so'ng `AdaptiveAvgPool2d(1)`, `Flatten`, `Linear(16,10)`. Bitta `F.cross_entropy` backward dan keyin tana chiqishining std, birinchi va oxirgi konvolyutsiya og'irlik gradienti normalarini (.3e) chop eting.
29. (*amaliy holat: tekstura klassifikatsiyasi*) `skimage.data.brick()`, `grass()`, `gravel()` (512×512 , sinflar 0, 1, 2) dan 32×32 lik qoplanmaydigan patchlar; yuqori yarmi (qator < 256) — train, pastki yarmi — test; har bir patch standartlanadi: $(p - \bar{p}) / (\text{std}(p) + 10^{-6})$. CNN: `Conv2d(1,8,5)`–`ReLU`–`MaxPool2d(2)`–`Conv2d(8,16,5)`–`ReLU`–`AdaptiveAvgPool2d(1)`–`Flatten`–`Linear(16,3)`; MLP: `Flatten`–`Linear(1024,64)`–`ReLU`–`Linear(64,3)`; `fit(..., epochs=20)`. To'plam shakllari, parametrlar, train/test accuracy va test confusion matrix (`sklearn.metrics.confusion_matrix`) ni chop eting.
30. (*amaliy holat: transfer learning*) Manba vazifa: Digits CNN (`n_out=5`) train'dagi 0–4 raqamlarida o'qitiladi. Maqsad vazifa: 5–9 raqamlari, har sinfdan faqat 10 ta belgilangan namuna (`rng = np.random.default_rng(42)`; har sinf $c = 5, \dots, 9$ uchun `rng.choice(np.where(ytr == c)[0], 10, replace=False)`), yoriqlar $y - 5$; test — test'dagi barcha 5–9. Uch variant, 100 epoxa (har biridan oldin `torch.manual_seed(42)`): noldan; manba modeli nusxasi (`copy.deepcopy`), barcha parametrlar muzlatilgan, yangi `Linear(128, 5)` bosh o'qitiladi; butun tarmoqni $\text{lr}=1e-3$ bilan fine-tuning. Manba test accuracy, har variant uchun o'qitiladigan parametrlar va maqsad test accuracy ni chop eting. Transfer bu yerda foyda berdimi? Nima uchun?